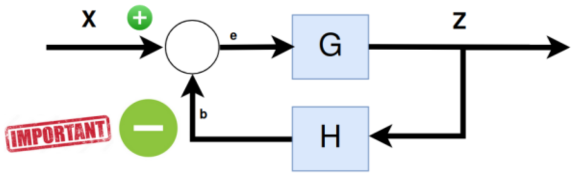
Retroalimentación negativa

Demuestra que $Z/X = \dfrac{G}{1+H\cdot G}$



Señal de error: $e = X - b = X - Z\cdot H$

$Z = G\cdot e$

luego $Z/G = X - Z\cdot H$

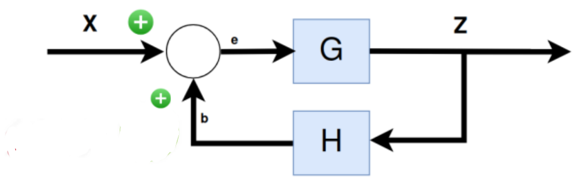
$Z\left(1/G + H\right) = X$

$Z\left(\dfrac{1+G\cdot H}{G}\right) = X$

$Z/X = \dfrac{G}{1+G\cdot H}$

Ejercicio 1: Demostrar que un sistema retroalimentado positivamente tiene la ecuación

$$\mathbf{Z} = \frac{\mathbf{G}}{\mathbf{1 - G \cdot H}} \cdot \mathbf{X}$$

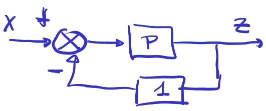


$e = Z\cdot H + X$ $\qquad Z = G\cdot e$

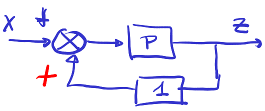
$Z/G = Z\cdot H + X$

$Z\left(\dfrac{1}{G} - H\right) = X \rightarrow Z/X = \dfrac{G}{1-H\cdot G}$

Ejercicio 2: Dibuja el diagrama de bloques de estas funciones de transferencia

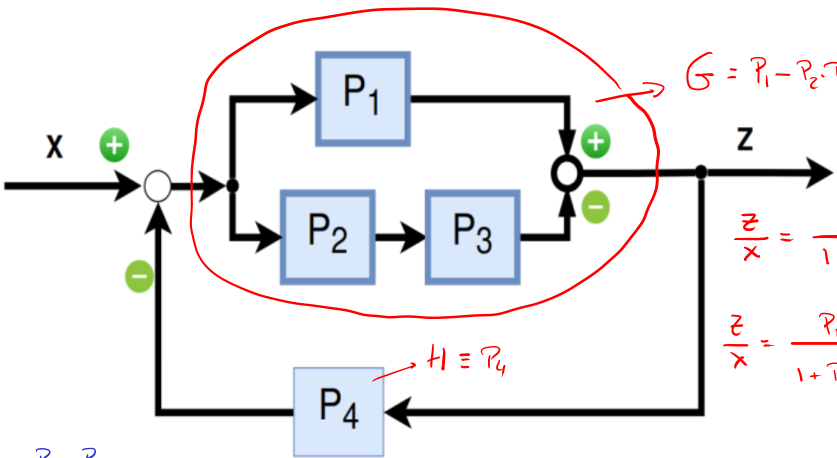


$$\mathbf{Z} = \frac{\mathbf{P}}{\mathbf{1+P}} \cdot \mathbf{X}$$



$$\mathbf{Z} = \frac{\mathbf{P}}{\mathbf{1-P}} \cdot \mathbf{X}$$

Ejercicio 3: Obtén la función de transferencia de este sistema



$G = P_1 - P_2\cdot P_3$

$H = P_4$

$\dfrac{Z}{X} = \dfrac{G}{1+H\cdot G}$

$\dfrac{Z}{X} = \dfrac{P_1-P_2\cdot P_3}{1+P_4\cdot(P_1-P_2\cdot P_3)}$

$Z = eP_1 - eP_2\cdot P_3$

$Z = e(P_1 - P_2 P_3)$

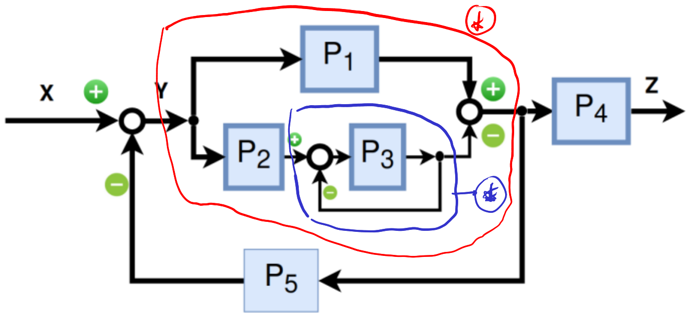
$e = X - Z P_4$

$\dfrac{Z}{P_1 - P_2\cdot P_3} = X - Z P_4$

$Z\left(\dfrac{1}{P_1-P_2\cdot P_3} + P_4\right) = X$

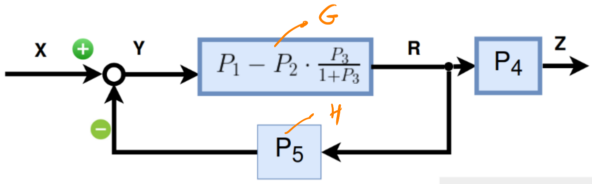
$Z\left(\dfrac{1+P_4(P_1-P_2\cdot P_3)}{P_1-P_2\cdot P_3}\right) = X$

$Z = \dfrac{P_1 - P_2\cdot P_3}{1+P_4(P_1-P_2\cdot P_3)}\, X$



(*) $\dfrac{P_3}{1+P_3}$

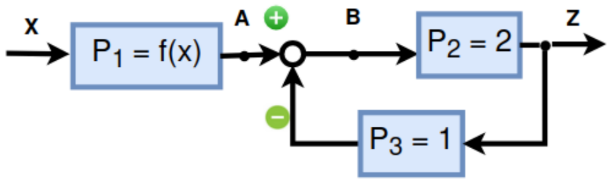
(✱) $P_1 - P_2 \cdot \dfrac{P_3}{1+P_3}$

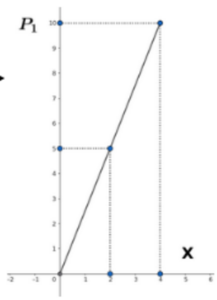


$$R/X = \frac{G}{1+G\cdot H} = \frac{P_1 - P_2 \frac{P_3}{1+P_3}}{1 + P_5\left(P_1 - P_2 \frac{P_3}{1+P_3}\right)}$$

$$\frac{Z}{X} = \frac{P_4 \cdot \left(P_1 - P_2 \frac{P_3}{1+P_3}\right)}{1 + P_5\left(P_1 - P_2\, P_3/_{1+P_3}\right)}$$

Ejercicio 5: Se sigue un sistema de control como el de la figura, siendo $P_1 = f(x)$





Calcular (a) El valor de la señal en A, B y Z cuando X = 4

Calcular (b) Calcular X cuando Z = 20 / 3

a) si $x = 4$ $P_1 = f(4) = 10$.

$$Z/A = \frac{P_2}{1+P_2\cdot P_3} = \frac{2}{1+2\cdot 1} = 2/3$$

$A = 10\cdot x$

$Z/X = 10 \cdot 2/3 = 20/3$

$Z = 20/3 \cdot 4 = 80/3$

$A = P_1 \cdot x = f(4)\cdot 4 = 40$

$B = A - Z\cdot P_3 = 40 - \frac{80}{3} = \frac{40}{3}$

b) $P_1 = m\cdot x$ siendo $m = 5/2$ (ya que $m\cdot 2 = 5$ ó $m \cdot 4 = 10$)

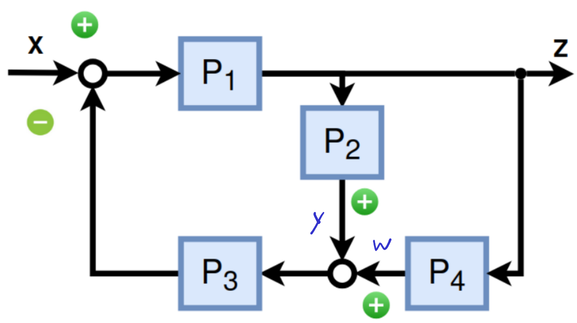
$P_1 = 5/2\, x$

$Z/A = 2/3$ si $A = P_1 \cdot x = 5/2\, x \cdot x = 5/2\, x^2$

$Z = 2/3 \cdot 5/2\, x^2 = 5/3\, x^2$ si $Z = 20/3$ $\quad 20/3 = 5/3\, x^2 \quad x^2 = 20/5 = 4$

$x = \pm 2$

Ejercicio 6: Obtener la función de transferencia del sistema Z=f(X)



$z = P_1 \cdot e \qquad e = x - (w+y)P_3$

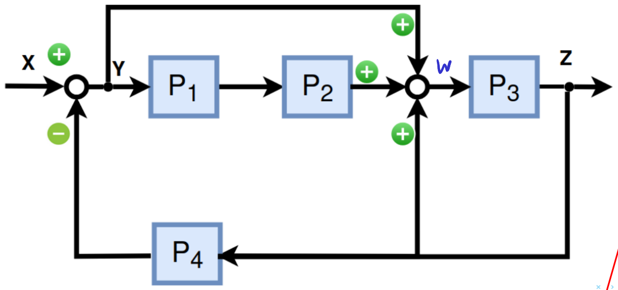
$y = P_2 \cdot z \,//\, w = P_4 \cdot z \rightarrow$

$e = x - (P_2+P_4)P_3\, z = z/P_1$

$$x = z\left(1/P_1 + (P_2+P_4)P_3\right) = z\,\frac{1+P_1(P_2+P_4)P_3}{P_1}$$

$$\frac{Z}{X} = \frac{P_1}{1 + P_1\cdot P_3\cdot(P_2+P_4)}$$

Ejercicio 7: En el siguiente ejercicio calcular Z = f(Y) y Z = f(x)



$y = x - P_4 \cdot Z$ $\qquad Z = P_3 \cdot W$

$W = Y + Y(P_1 \cdot P_2) + Z = Z/P_3$

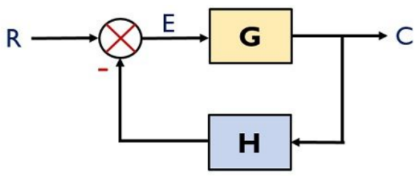
$Y(1 + P_1P_2) = Z\left(1/P_3 - 1\right) = Z\,\frac{1-P_3}{P_3}$

$$Z/Y = \frac{(1+P_1P_2)P_3}{1-P_3}$$

$$Y/Z = \frac{1-P_3}{(1+P_1P_2)P_3} = \frac{X}{Z} - P_4 \Rightarrow X/Z = P_4 + \frac{1-P_3}{(1+P_1P_2)P_3} = \frac{P_3P_4(1+P_1P_2)+(1-P_3)}{(1+P_1P_2)P_3}$$

$$Z/X = \frac{(1+P_1P_2)P_3}{P_3P_4(1+P_1P_2)+1-P_3}$$

Ejercicio 8: ¿Puede representarse el resultado Z=f(X) del ejercicio 7 por un sistema simple realimentado negativamente?



$$\frac{C}{R} = \frac{G}{1+GH}$$

Divido todo entre $(1-P_3)$

$$Z/X = \frac{\overbrace{(1+P_1P_2)P_3/(1-P_3)}^{G}}{1 + \underbrace{P_4}_{H}\,\overbrace{\frac{P_3(1+P_1P_2)}{1-P_3}}^{G}}$$

Sí $G = \frac{(1+P_1P_2)P_3}{1-P_3}$

$H = P_4$

Sí, puedo hacerlo.

O bien si $$Z/X = \frac{(1+P_1P_2)P_3}{P_3P_4(1+P_1P_2)+1-P_3} = \frac{1}{P_4 + \frac{1-P_3}{P_3(1+P_1P_2)}} = \frac{1/P_4}{1 + \frac{1}{P_4}\,\frac{1-P_3}{P_3(1+P_1P_2)}}$$

$G = 1/P_4$ $\qquad H = \frac{1-P_3}{P_3(1+P_1P_2)}$